

## Direct On Line Forward - Reverse Motor MAPS Template (Electrical)

The Direct On Line Forward - Reverse Motor MAPS template is designed to control a motor that is connected as a DOL, which is controlled by two outputs: one for forward direction and another for the reverse direction.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This document describes the Advanced Direct On Line Forward - Reverse Motor MAPS template. For details of its required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

### Advanced DOL Forward - Reverse Motor Functional Philosophy

The DOL forward and reverse direction outputs are enabled once all interlocks are ready and a start request is received.

#### Field Interlocks

- MCC (Overload) Trip Feedback
- Isolator On Feedback
- Safety Interlock Feedback
- Motor Current Feedback (Analog Input) (Current above High Limit Setpoint)

#### PLC Interlocks

- Start Interlock Ready
- Process Interlock Ready
- Restart Delay Interlock
- There are no interlock faults latched

**Motor Current** – The motor current is monitored constantly after the running setpoint time has elapsed, if the current goes above the high current limit for five seconds the motor will trip on overcurrent.

**Interlock fault recovery process** – once a field interlock is activated the faceplate will display a red indication, if the interlock is recovered and still latched in the PLC function block and the faceplate will show a green indication and the reset button on the faceplate will flash, pressing on the reset button will unlatch the interlock and display a grey indication and allow the DOL to become ready for operation.

**Maintenance mode** - will disable the Start and Process Interlocks, but respects all the other interlocks. This is used to test a DOL while the process is stopped.

**Simulation mode** - will disable the physical output of the DOL and simulate an input run feedback after the elapsed setpoint time from the faceplate. The Start and Process Interlocks are respected. Simulation mode is used to test the process logic without starting the physical device itself. It is very useful in testing, commissioning and maintenance.

**Start and stop requests are initiated by selecting a mode on the DOL faceplate:**

- **Auto mode** - the auto control program initiates a start and stop, normal PLC control.
- **Manual mode & Desk mode** - the operator can start and stop from the faceplate.
- **Manual mode & Field mode** - the motor can be started and stopped from the start and stop push buttons in the field.

## Typical Faceplate Graphics for the Advanced DOL Forward - Reverse Motor MAPS Template

| DOL ISO          | DOL MOTOR LEFT                   | DOL MOTOR RIGHT | DOL PUMP LEFT | DOL PUMP RIGHT |
|------------------|----------------------------------|-----------------|---------------|----------------|
|                  |                                  |                 |               |                |
|                  | DOL Reverse Direction Indication |                 |               |                |
|                  | DOL Forward Direction Indication |                 |               |                |
|                  | Manual Mode                      |                 |               |                |
|                  | Maintenance Mode                 |                 |               |                |
|                  | Simulation Mode                  |                 |               |                |
|                  | Motor Control Name               |                 |               |                |
| Graphics legend: |                                  |                 |               |                |
| Line colour      | Fill colour                      | DOL status      |               |                |
| Black            | Grey                             | Stopped         |               |                |
| Grey             | White                            | Stopping        |               |                |
| Green            | White                            | Starting        |               |                |
| Black            | Green                            | Running         |               |                |
| Black            | Yellow/Red                       | Fault Active    |               |                |

| Signal Description          |                                     | Agent Type | Advanced | Standard | Basic |
|-----------------------------|-------------------------------------|------------|----------|----------|-------|
| <b>SCADA Control Words:</b> |                                     |            |          |          |       |
| SSCL 0                      | Control Word                        | Marshal    | x        |          |       |
| SSCL 1                      | Fail To Start Setpoint              | Analog     | x        |          |       |
| SSCL 2                      | Fail To Stop Setpoint               | Analog     | x        |          |       |
| SSCL 3                      | Running Setpoint                    | Analog     | x        |          |       |
| SSCL 4                      | Stopping Setpoint                   | Analog     | x        |          |       |
| SSCL 5                      | Simulate Setpoint                   | Analog     | x        |          |       |
| SSCL 6                      | Restart Setpoint                    | Analog     | x        |          |       |
| SSCL 7                      | Max Current Setpoint (High Limit)   | Analog     | x        |          |       |
| SSCL 8                      | Min In Current Setpoint (Raw Min)   | Analog     | x        |          |       |
| SSCL 9                      | Max In Current Setpoint (Raw Max)   | Analog     | x        |          |       |
| SSCL 10                     | Min Out Current Setpoint (Eng. Min) | Analog     | x        |          |       |
| SSCL 11                     | Max Out Current Setpoint (Eng. Max) | Analog     | x        |          |       |

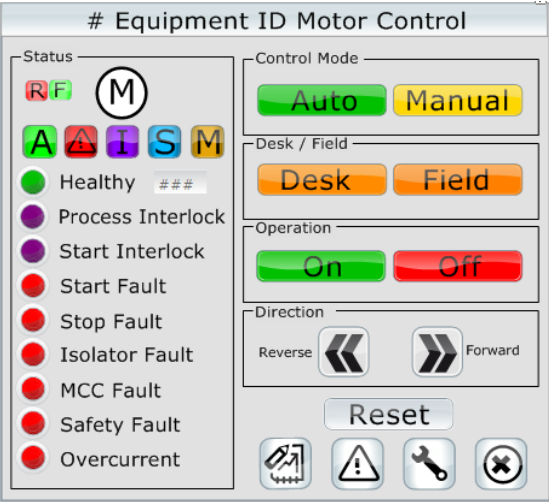
| Signal Description         |                                       | Agent Type | Advanced | Standard | Basic |
|----------------------------|---------------------------------------|------------|----------|----------|-------|
| <b>SCADA Status Words:</b> |                                       |            |          |          |       |
| SSSL 0                     | Status Word                           | Marshal    | x        |          |       |
| SSSL 1                     | Starting Time Process Value (actual)  | Analog     | x        |          |       |
| SSSL 2                     | Stopping Time Process Value (actual)  | Analog     | x        |          |       |
| SSSL 3                     | No of Operations (actual)             | Analog     | x        |          |       |
| SSSL 4                     | Restart Delay Time Remaining (actual) | Analog     | x        |          |       |
| SSSL 5                     | Motor Current Process Value (actual)  | Analog     | x        |          |       |

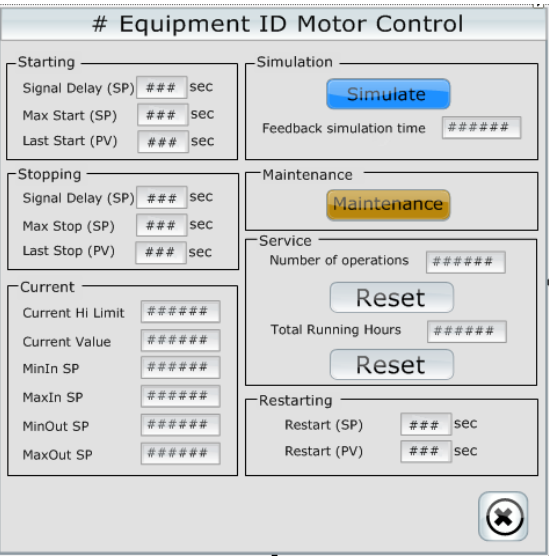
|                         |                                         |        |                 |                 |              |
|-------------------------|-----------------------------------------|--------|-----------------|-----------------|--------------|
| SSSL 6                  | Total Running Hours                     | Analog | x               |                 |              |
| <b>Digital Inputs:</b>  |                                         |        | <b>Advanced</b> | <b>Standard</b> | <b>Basic</b> |
| DI 0                    | Running Signal Forward                  |        | x               |                 |              |
| DI 1                    | Running Signal Reverse                  |        | x               |                 |              |
| DI 2                    | Isolator Healthy (On = Healthy)         |        | x               |                 |              |
| DI 3                    | MCC Healthy (On = Healthy)              |        | x               |                 |              |
| DI 4                    | Safety Interlock Healthy (On = Healthy) |        | x               |                 |              |
| DI 5                    | Manual Start Push Button Forward        |        | x               |                 |              |
| DI 6                    | Manual Start Push Button Reverse        |        | x               |                 |              |
| DI 7                    | Manual Stop Push Button (Off = Stop)    |        | x               |                 |              |
| <b>Digital Outputs:</b> |                                         |        |                 |                 |              |
| DO 0                    | Device Start Forward                    |        | x               |                 |              |
| DO 1                    | Device Start Reverse                    |        | x               |                 |              |
| <b>Analog Input:</b>    |                                         |        |                 |                 |              |
| AI 0                    | Analog Input for Current Drawn          |        | x               |                 |              |

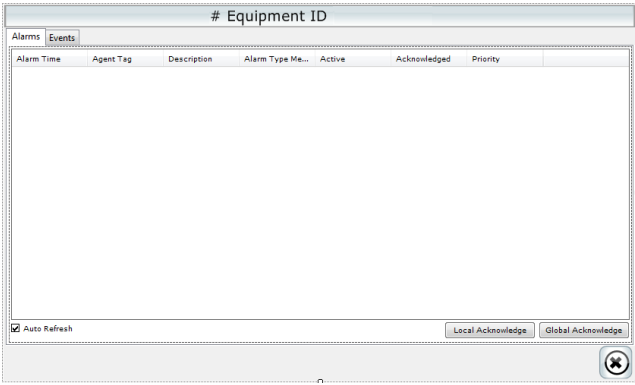
| SCADA Control Word (Adroit Marshal Agent) |                                 | Event Logged | Advanced | Standard | Basic |
|-------------------------------------------|---------------------------------|--------------|----------|----------|-------|
| Bit 0                                     | Auto/Manual                     | x            | x        |          |       |
| Bit 1                                     | Field/Desk                      | x            | x        |          |       |
| Bit 2                                     | Maintenance                     | x            | x        |          |       |
| Bit 3                                     | Simulation                      | x            | x        |          |       |
| Bit 4                                     | Alarms All Reset                |              | x        |          |       |
| Bit 5                                     | Reset No Operations Accumulator | x            | x        |          |       |
| Bit 6                                     | Start Command                   | x            | x        |          |       |
| Bit 7                                     | Reset Running Hours Accumulator | x            | x        |          |       |
| Bit 8                                     | Stop Command                    | x            | x        |          |       |
| Bit 9                                     | Spare                           |              |          |          |       |
| Bit 10                                    | Spare                           |              |          |          |       |
| Bit 11                                    | Spare                           |              |          |          |       |
| Bit 12                                    | Spare                           |              |          |          |       |
| Bit 13                                    | Spare                           |              |          |          |       |
| Bit 14                                    | Forward Command                 | x            | x        |          |       |
| Bit 15                                    | Reverse Command                 | x            | x        |          |       |

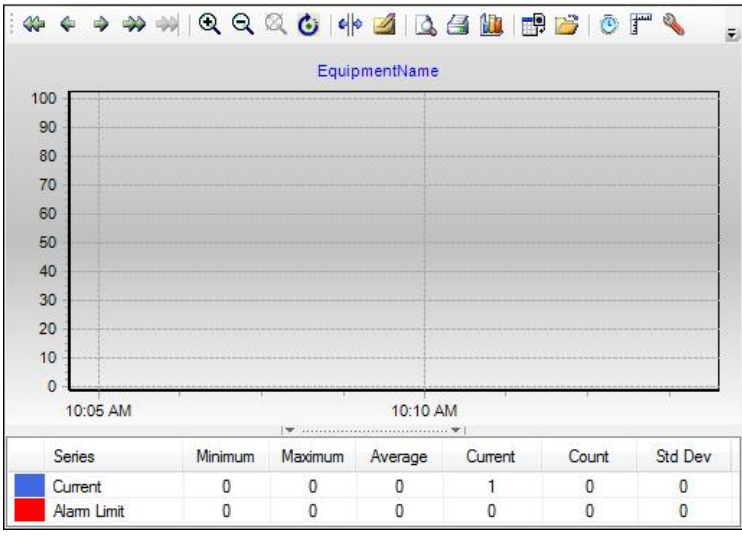
| SCADA Status Word (Adroit Marshal Agent) |                                   | Alarmed | Advanced | Standard | Basic |
|------------------------------------------|-----------------------------------|---------|----------|----------|-------|
| Bit 0                                    | Process Interlock                 |         | x        |          |       |
| Bit 1                                    | Ready To Start                    |         | x        |          |       |
| Bit 2                                    | Stopped                           |         | x        |          |       |
| Bit 3                                    | Starting                          |         | x        |          |       |
| Bit 4                                    | Running                           |         | x        |          |       |
| Bit 5                                    | Stopping                          |         | x        |          |       |
| Bit 6                                    | Fault                             |         | x        |          |       |
| Bit 7                                    | Start Interlock OK                |         | x        |          |       |
| Bit 8                                    | Isolator Closed Fault             | x       | x        |          |       |
| Bit 9                                    | Fault Fail To Stop                | x       | x        |          |       |
| Bit 10                                   | Fault Fail To Start               | x       | x        |          |       |
| Bit 11                                   | MCC Fault                         | x       | x        |          |       |
| Bit 12                                   | Safety Fault                      | x       | x        |          |       |
| Bit 13                                   | Overcurrent Fault                 | x       | x        |          |       |
| Bit 14                                   | Spare                             |         |          |          |       |
| Bit 15                                   | Forward Reverse Status Indication |         | x        |          |       |

| Template: Advanced DOL Forward - Reverse PLC Function Block |                          |                           |                                           |
|-------------------------------------------------------------|--------------------------|---------------------------|-------------------------------------------|
| <Device> (Advanced DOL F/R Control)                         |                          |                           |                                           |
|                                                             |                          | FB_DOL_FR_A.v1.0          |                                           |
|                                                             |                          | DOL_FR_A.v1.0             |                                           |
| DATA_DOL_FR_A_V1.0.F.I.Run_FWD_FB                           | F.I.Run_FWD_FB           | F.O_DOL_START_FWD_CMD     | DATA_DOL_FR_A_V1.0.F.O_DOL_START_FWD_CMD  |
| DATA_DOL_FR_A_V1.0.F.I.Run_REV_FB                           | F.I.Run_REV_FB           | F.O_DOL_START_REV_CMD     | DATA_DOL_FR_A_V1.0.F.O_DOL_START_REV_CMD  |
| DATA_DOL_FR_A_V1.0.F.I.Isolator_Closed_FB                   | F.I.Isolator_Closed_FB   | S.SW1_Marshall            | DATA_DOL_FR_A_V1.0.S.SW1_Marshall         |
| DATA_DOL_FR_A_V1.0.F.I.MCC_Healthy_FB                       | F.I.MCC_Healthy_FB       | S.SW2_Starting_Time_PV    | DATA_DOL_FR_A_V1.0.S.SW2_Starting_Time_PV |
| DATA_DOL_FR_A_V1.0.F.I.Safety_Interlock_OK                  | F.I.Safety_Interlock_OK  | S.SW3_Stopping_Time_PV    | DATA_DOL_FR_A_V1.0.S.SW3_Stopping_Time_PV |
| DATA_DOL_FR_A_V1.0.VAR.Process_Int_OK                       | VAR.Process_Int_OK       | S.SW4_No_Of_Operations    | DATA_DOL_FR_A_V1.0.S.SW4_No_Of_Operations |
| DATA_DOL_FR_A_V1.0.VAR.Start_Int_OK                         | VAR.Start_Int_OK         | S.SW5_Restart_PV          | DATA_DOL_FR_A_V1.0.S.SW5_Restart_PV       |
| DATA_DOL_FR_A_V1.0.VAR.Auto_Run_FWD_Req                     | VAR.Auto_Run_FWD_Req     | S.SW6_Current_PV          | DATA_DOL_FR_A_V1.0.S.SW6_Current_PV       |
| DATA_DOL_FR_A_V1.0.VAR.Auto_Run_REV_Req                     | VAR.Auto_Run_REV_Req     | S.SW7_Running_Hours       | DATA_DOL_FR_A_V1.0.S.SW7_Running_Hours    |
| DATA_DOL_FR_A_V1.0.F.I.MAN_START_FWD_PB                     | F.I.MAN_START_FWD_PB     | S.CW1_0_Auto_Manual       |                                           |
| DATA_DOL_FR_A_V1.0.F.I.MAN_START_REV_PB                     | F.I.MAN_START_REV_PB     | S.CW1_1_Desk_Field        |                                           |
| DATA_DOL_FR_A_V1.0.F.I.MAN_STOP_PB                          | F.I.MAN_STOP_PB          | S.CW1_2_Maintenance       |                                           |
| DATA_DOL_FR_A_V1.0.S.CW1_Marshall                           | S.CW1_Marshall           | S.CW1_3_Simulation        |                                           |
| DATA_DOL_FR_A_V1.0.S.CW2_Fail_To_Start_SP                   | S.CW2_Fail_To_Start_SP   | S.SW1_1_Ready_To_Start    |                                           |
| DATA_DOL_FR_A_V1.0.S.CW3_Fail_To_Stop_SP                    | S.CW3_Fail_To_Stop_SP    | S.SW1_2_Stopped           |                                           |
| DATA_DOL_FR_A_V1.0.S.CW4_Running_SP                         | S.CW4_Running_SP         | S.SW1_3_Starting_RQ       |                                           |
| DATA_DOL_FR_A_V1.0.S.CW5_Stopping_SP                        | S.CW5_Stopping_SP        | S.SW1_4_Running           |                                           |
| DATA_DOL_FR_A_V1.0.S.CW6_Simulate_SP                        | S.CW6_Simulate_SP        | S.SW1_5_Stopping_RQ       |                                           |
| DATA_DOL_FR_A_V1.0.S.CW7_Restart_SP                         | S.CW7_Restart_SP         | S.SW1_9_Fault_Stop        |                                           |
| DATA_DOL_FR_A_V1.0.S.CW8_MaxCurrent_SP                      | S.CW8_MaxCurrent_SP      | S.SW1_A_Fault_Start       |                                           |
| DATA_DOL_FR_A_V1.0.S.CW9_MinIn_Current_SP                   | S.CW9_MinIn_Current_SP   | S.SW1_D_Fault_OverCurrent |                                           |
| DATA_DOL_FR_A_V1.0.S.CW10_MaxIn_Current_SP                  | S.CW10_MaxIn_Current_SP  |                           |                                           |
| DATA_DOL_FR_A_V1.0.S.CW11_MinOut_Current_SP                 | S.CW11_MinOut_Current_SP |                           |                                           |
| DATA_DOL_FR_A_V1.0.S.CW12_MaxOut_Current_SP                 | S.CW12_MaxOut_Current_SP |                           |                                           |
| DATA_DOL_FR_A_V1.0.F.AI.Current                             | F.AI.Current_PV          |                           |                                           |

| Template Graphic: Advanced Home Screen                                           | Control           | Descriptions                                               |
|----------------------------------------------------------------------------------|-------------------|------------------------------------------------------------|
|  | <b>Auto</b>       | Auto control by PLC                                        |
|                                                                                  | <b>Manual</b>     | Manual control by operator, desk and field mode            |
|                                                                                  | <b>Desk</b>       | Manual control from the SCADA faceplate                    |
|                                                                                  | <b>Field</b>      | Manual control from the field start/stop station           |
|                                                                                  | <b>On</b>         | Manual start the DOL                                       |
|                                                                                  | <b>Off</b>        | Manual stop the DOL                                        |
|                                                                                  | <b>Reset</b>      | Reset the DOL after a fault has been repaired              |
|                                                                                  | Healthy ###       | Healthy to start, ### - Restart delay countdown in seconds |
|                                                                                  | Process Interlock | DOL process ok to run interlock ready                      |
|                                                                                  | Start Interlock   | DOL ok to start interlock ready                            |
|                                                                                  | Start Fault       | DOL failed to start fault                                  |
|                                                                                  | Stop Fault        | DOL failed to stop fault                                   |
|                                                                                  | Isolator Fault    | DOL isolator off fault                                     |
|                                                                                  | MCC Fault         | DOL MCC tripped fault                                      |
|                                                                                  | Safety Fault      | DOL Safety off fault                                       |
|                                                                                  | Overcurrent       | DOL overcurrent fault                                      |
|                                                                                  | Reverse           | DOL reverse direction selection and indication             |
|                                                                                  | Forward           | DOL forward direction selection and indication             |

| Template Graphic: Advanced Setup Screen                                           | Control                 | Descriptions                                                  |
|-----------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------|
|  | <b>Simulate</b>         | Enable simulation of DOL, disables the physical output        |
|                                                                                   | Feedback Time           | Time for simulation to simulate run feedback signal           |
|                                                                                   | <b>Maintenance</b>      | Enable maintenance mode (Disables all interlocks)             |
|                                                                                   | Number of operations    | Total number of DOL start operations                          |
|                                                                                   | Running Hours           | Total number of DOL running hours                             |
|                                                                                   | <b>Reset</b>            | Reset the total counter of the current service                |
|                                                                                   | Start Signal Delay (SP) | Delay the PLC to accept the DOL is running                    |
|                                                                                   | Max Start (SP)          | Time for running signal is on after start signal is on        |
|                                                                                   | Last Start (PV)         | Last time for running signal was on after start signal was on |
|                                                                                   | Stop Signal Delay (SP)  | Delay the PLC to accept the DOL is stopped                    |
|                                                                                   | Max Stop (SP)           | Time running signal is off after start signal is off          |
|                                                                                   | Last Stop (PV)          | Last time running signal was off after start signal was off   |
|                                                                                   | Current Hi Limit        | Sets the value of the maximum current for the DOL to trip at  |
|                                                                                   | Current Value           | Actual current the DOL is running at                          |
|                                                                                   | MinIn, MaxIn            | Raw current input values for scaling                          |
|                                                                                   | MinOut, MaxOut          | Engineering current values for scaling                        |
|                                                                                   | Restart (SP)            | Restart prevention time of the DOL setpoint                   |
|                                                                                   | Restart (PV)            | Remaining time before a DOL restart is allowed                |

| Template Graphic: Advanced Alarms & Events Screen                                 | Control            | Descriptions                                                    |
|-----------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------|
|  | Alarms             | Display all the alarms filtered for the current Digital Output  |
|                                                                                   | Events             | Display all the events filtered for the current Digital Output. |
|                                                                                   | Local Acknowledge  | Acknowledge all the alarms on locally on the current machine    |
|                                                                                   | Global Acknowledge | Acknowledge all the alarms on globally on all the machines.     |

| Template Graphic: Advanced Trend Screen                                            | Control     | Descriptions                                                                                                                |
|------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------|
|  | Current     | Display the current of the DOL in the trend, with minimum, maximum and average values.                                      |
|                                                                                    | Alarm Limit | Display the current Setpoint of alarming on high current of the DOL in the trend, with minimum, maximum and average values. |